

# Project Report: Machine Learning Pipeline for Adverse Drug Reaction Severity Prediction

## 1. Problem Understanding

### 1.1 What are Adverse Drug Reactions (ADRs)?

Adverse Drug Reactions (ADRs) are unwanted, harmful, or unintended side effects that occur when a patient takes a medication at normal, recommended therapeutic doses. ADRs represent a significant clinical challenge, distinct from accidental overdoses or drug misuse, as they occur under standard compliance and prescribing guidelines.

### 1.2 Clinical Significance of ADR Prediction

Predicting ADRs is crucial in clinical workflows because:

- **Patient Safety:** Enables proactive risk assessment to prevent severe reactions, organ damage, or death.
- **Clinical Decision Support:** Assists clinicians in identifying high-risk patients and adjusting dosages or selecting alternative therapies before prescribing.
- **Healthcare Costs:** Reduces the frequency of emergency admissions, length of hospital stays, and overall clinical burden.

### 1.3 Machine Learning Formulation

This problem is formulated as a **Binary Classification** task where:

- **Input Features (\$X\$):** Patient demographics (age, gender), reporting country, and drug dosage forms (tablet, capsule, injection, etc.).
- **Target Label (\$y\$):** ADR Severity, classified as **Serious (1)** (results in hospitalization, death, or life-threatening situations) or **Non-Serious (0)**.

Decision Trees are highly suitable for this domain because:

1. **Interpretability:** Doctors and regulatory authorities require transparent clinical logic. Decision trees provide clear, human-readable "if-then" rules.
  2. **Data Flexibility:** They naturally handle mixed data types (numerical age, categorical countries/forms) without requiring feature scaling (like normalization).
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## 2. Preprocessing & Data Cleaning

Healthcare datasets are notoriously noisy, inconsistent, and suffer from missing records. We implemented the following preprocessing steps:

1. **Missing Value Imputation:**
  - Patient **age** was imputed with the **median** (49 years) to protect against skewness

from outliers.

- gender was imputed with code '0' (representing **Unknown**).
  - drug\_dosage\_form and country missing values were filled with the category 'Unknown'.
2. **Standardization of Dosage Forms:** Raw text forms from openFDA contain typos and inconsistent naming. We mapped them to six clean categories: Tablet, Capsule, Injection, Oral Solution, Suspension, and Other.
  3. **Age Binning:** Created five categorical age cohorts: Pediatric (<18), Young Adult (18-35), Adult (36-50), Middle Aged (51-65), and Senior (>65).
  4. **Target Encoding:** Converted adr\_severity (where 1 = Serious, 2 = Non-Serious) to binary targets 1 (Serious) and 0 (Non-Serious).
  5. **Feature Encoding:** Applied One-Hot Encoding to categorical variables (gender\_name, dosage\_form\_clean, country\_clean) to prepare them for Scikit-Learn.
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### 3. Exploratory Data Analysis (EDA) - Key Insights

- **Demographics:** Patients reporting ADRs are predominantly middle-aged and senior, with the distribution peaking around 50–70 years. Females slightly outnumber males in reports.
  - **Severity Distribution:** A high ratio of reported adverse events is classified as serious, which is typical for post-market surveillance databases because patients and doctors rarely report minor, self-resolving symptoms.
  - **Clinical Outcome Flag Correlations:** Serious outcomes, specifically hospitalization and death, show a strong positive correlation with adr\_severity. Age has a mild positive correlation, demonstrating that older cohorts face higher risks of severe reactions.
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### 4. Modeling & Hyperparameter Tuning

We split our preprocessed dataset into **80% training** and **20% testing** sets, maintaining stratified target proportions.

#### 4.1 Baseline Decision Tree

A baseline Decision Tree Classifier was trained with default settings.

- **Maximum Depth:** 27
- **Leaf Nodes:** 176
- **Characteristics:** This tree is highly complex, deep, and overfitted to the training noise.

## 4.2 Tuned Decision Tree (GridSearchCV)

We used 5-fold cross-validation over a parameter grid to optimize the model:

- **Grid Parameters:** max\_depth (3, 5, 8, 10, None), min\_samples\_split (2, 5, 10, 20), min\_samples\_leaf (1, 2, 4, 8), and criterion ('gini', 'entropy').
  - **Best Hyperparameters:** {'criterion': 'gini', 'max\_depth': 5, 'min\_samples\_leaf': 8, 'min\_samples\_split': 2}
  - **Optimized Tree Depth:** 5
  - **Optimized Leaf Nodes:** 22
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## 5. Model Evaluation & Comparison

We evaluated the baseline Decision Tree, tuned Decision Tree, and a Random Forest Classifier on the unseen test set:

| Evaluation Metric | Baseline Decision Tree | Tuned Decision Tree | Random Forest Classifier |
|-------------------|------------------------|---------------------|--------------------------|
| Accuracy          | 0.8350                 | 0.8350              | 0.8350                   |
| Precision         | 0.8713                 | 0.8713              | 0.8713                   |
| Recall            | 0.7720                 | 0.7720              | 0.7720                   |
| F1-score          | 0.8187                 | 0.8187              | 0.8187                   |

### Why F1-Score is Crucial in Imbalanced Healthcare Data

F1-score is the harmonic mean of Precision and Recall. In imbalanced datasets (e.g., if 95% of patients are healthy and 5% have a disease), a model can achieve 95% accuracy by simply predicting 'healthy' for everyone. However, its Recall for the diseased class would be 0%. F1-score prevents this false sense of security by penalizing models that fail to capture both Precision and Recall.

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## 6. Feature Importance & Clinical Interpretation

Our tuned model extracted the following feature importance weights:

1. **country\_clean\_US:** 46.3%
2. **hospitalization:** 42.4%
3. **death:** 10.7%
4. **age:** 0.5%
5. **Other features (gender, dosage form):** 0.0%

## Interpretation

Clinical outcome indicators like `hospitalization` and `death` dominate the classification rules. Under FDA guidelines, any adverse event that results in hospitalization or death is automatically categorized as a serious event. This direct medical mapping explains why the models achieve identical performance; they are all picking up on these dominant logical rules.

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## 7. Conclusions & Healthcare Challenges

- **Challenges in Healthcare Data Cleaning:** Mismatched clinic entries, free-text typos in drug dosage names, and incomplete demographic records require robust standardization pipelines (such as keyword matching) to prevent data corruption.
- **Mitigating Tree Overfitting:** Grid search and pre-pruning (restricting `max_depth` and requiring minimum leaf samples) are essential. Pruning reduced our tree depth from 27 to 5, preventing it from memorizing noise and ensuring the decision rules remain interpretable for medical personnel.
- **Random Forest Advantage:** In more complex, less deterministic datasets, Random Forest outperforms single decision trees by combining multiple decorrelated trees, averaging out individual tree variances to improve generalization.